

AGUWA, CHINEDU

Kubwa, Abuja, FCT, Nigeria

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EDUCATION

Federal University of Technology, Minna, Nigeria.

Mar 2019 - Jan 2025

Bachelor of Engineering (B.Eng.) in Civil Engineering

CGPA: 4.79/5.0 (Scholaro Evaluation: 3.89/4.0) | First Class Honours | Top 2 of the department

WORK EXPERIENCE

Environmental Sustainability Software Engineer (*Part-time*)

Apr 2025 – Apr 2026

EDAT Climate Data and Analytics — Houston, Texas (Remote)

- Led backend architecture and development of a production-grade carbon accounting platform supporting Scope 1, 2, and 3 emissions tracking across stationary combustion, mobile combustion, fugitive emissions, purchased electricity, renewable energy, waste (EPA WARM v16, 61 material types, 6 pathways), and water modules.
- Designed and implemented Django ORM data models for 10+ activity types, each linked to a centralised emission record system with version-controlled GWP conversion outputs (AR5 100-year basis) storing CO₂, CH₄, N₂O, and CO₂e in kg.
- Engineered asynchronous task pipelines using Django and Celery for large-scale emissions batch processing, GWP computations, and scheduled reporting workflows.
- Integrated AI-driven emissions classification and validation services into the reporting pipeline, improving data quality and reducing manual review overhead.
- Collaborated on frontend implementation using Next.js and ShadCN UI, delivering interactive data visualisation dashboards for Scope-level emissions breakdowns and trend analysis.
- Conducted independent research across EPA, DEFRA, and IPCC emission factor methodologies to inform calculation logic across all four accounting modules (emissions, energy, waste, water).

Full Stack Developer (*3-month contract*)

Dec 2024 – Mar 2025

Eagle AI Labs — Dubai, UAE (Remote)

- Designed and implemented scalable REST APIs using NestJS and PostgreSQL for a crypto token platform, supporting high-throughput transaction processing and secure data management.
- Engineered tokenomics logic including rewards distribution systems, token vesting schedules, and secure wallet management modules handling multi-currency operations.
- Integrated Retrieval-Augmented Generation (RAG) based AI features for intelligent user assistance alongside multiple third-party crypto market APIs (price feeds, portfolio tracking) into a unified analytics dashboard.
- Delivered complete backend-to-dashboard feature set within a 3-month contract timeline, collaborating with a cross-functional remote team across multiple time zones.

Mobile Developer

Feb 2024 – Nov 2024

SureData Consulting Ltd — Tilbury, UK (Remote)

- Architected and shipped ChapelMate, a production-grade church management system for iOS and Android using React Native and Expo, successfully published on both the Google Play Store and Apple App Store, serving real end users across multiple church communities.
- Contributed to the ChapelMate backend on Ubuntu servers using Node.js, Express, and PostgreSQL, implementing JWT authentication, RESTful API architecture, relational database schema design, and real-time data synchronisation.
- Implemented contribution tracking, Paypal, and push notification systems, enabling church administrators to manage memberships and finances within a single platform.
- Delivered feature enhancements for CallUp247, a Flutter-based business communication app published on the Google Play Store and App Store, supporting customers to service-providers interaction flows and real-time messaging.
- Contributed core mobile features and API integrations to Kint, a cross-platform health management application built with Flutter, supporting patient data workflows and third-party health API connectivity.

Mobile & Full Stack Developer Intern

Nov 2023 – Jan 2024

Exinn Digital Technology — Maputo, Mozambique (Remote)

- Developed a full-stack medical web application to support nursing mothers, built with TypeScript, React, Node.js, NestJS, Supabase, and MongoDB, covering user authentication, clinical content delivery, and personalised health guidance.
- Built and integrated a patient-facing mobile application using React Native, embedding an AI-powered chatbot for doctor-patient interaction and a clinical note generator leveraging Google Gemini generative AI models to reduce documentation time for healthcare professionals.

Software Engineering Trainer (Part-time)

Jun 2023 – Oct 2023

A-Z New Age Tutor — Nigeria (On-site)

- Designed and delivered a structured practical training curriculum for 5 undergraduate software engineering students, covering Python, JavaScript, React, Node.js, and introductory machine learning applications across a 4-month programme.
- Developed hands-on project-based exercises to reinforce core programming concepts, resulting in students completing functional end-to-end projects by programme completion.

Software Developer Intern

Aug 2021 – Jun 2023

Scholarly Pathways — Nigeria (Remote)

- Designed, developed, and deployed a WordPress-based scholarship news and updates platform, implementing full SEO optimisation via Google Search Console and search indexing to drive organic traffic growth.
- Integrated Google AdSense monetisation, enabling the platform to generate revenue while serving scholarship discovery content to undergraduate students across Nigeria.
- Maintained and iterated on the platform over a 22-month engagement, managing content architecture, plugin configuration, and performance optimisation.

SKILLS

Languages: Python · JavaScript · TypeScript · Dart · SQL · Go

Frontend: React · Next.js · Tailwind CSS · ShadCN UI · Bootstrap · Sass

Backend: Django · Django REST Framework · Node.js · Express · NestJS · FastAPI · Celery

Mobile: React Native · Expo · Flutter · TensorFlow Lite (*on-device ML*)

Databases & Storage: PostgreSQL · MongoDB · MySQL · Firebase · Supabase · Neon

AI & Machine Learning: TensorFlow · Scikit-Learn · Pandas · NumPy · OpenCV · MATLAB · Random Forest · CNN · RAG pipelines · Gemini AI

Cloud, DevOps & Tools: Docker · AWS · Azure · Ubuntu Server · CI/CD · Git · GitHub · Joblib

Domain Expertise: Carbon accounting · EPA WARM v16 · DEFRA & IPCC emission factors · Lifecycle Assessment (LCA) · Embodied carbon quantification · Structural engineering & BIM · Geotechnical investigation

RECENT PERSONAL PROJECTS

BuildCore — Sustainable Construction Management Platform

Next.js · Django · React Native · PostgreSQL ·

- Architecting a full-stack construction management platform integrating Bills of Materials (BoM), material inventory tracking, wastage management, and embodied carbon accounting into a unified project reporting system.
- Built a Django REST Framework backend with modular APIs supporting real-time material consumption logging, wastage quantification, and project-level carbon emission calculations aligned with lifecycle assessment (LCA) principles.
- Developing a cross-platform mobile companion app in React Native for on-site material tracking, enabling field engineers to log inventory movements, flag wastage events, and sync data in real time with the web dashboard.
- Implementing an AI-driven material recommendation engine to suggest lower-carbon construction material substitutes based on performance, cost, and environmental constraints, contributing toward net-zero construction decision support.

Pavement Distress Detection System

TensorFlow · Flutter · FastAPI · Google Maps SDK

- Designed and trained a Convolutional Neural Network (CNN) on 30,000 labelled pavement images to classify multiple distress types (cracks, potholes) with confidence-scored predictions, exported to TensorFlow Lite for on-device inference.
- Built a full-stack mobile application in Flutter integrating the TensorFlow Lite model with real-time GPS tagging, detection logging, and Google Maps visualisation of historical distress locations across road networks.
- Implemented an image preprocessing pipeline using OpenCV (resizing to 224×224, normalisation, channel formatting) to optimise model accuracy and inference speed on resource-constrained mobile devices.
- Developed a FastAPI backend handling user authentication, detection history storage, geospatial analytics, and an admin dashboard for centralised network-wide distress review, enabling scalable infrastructure monitoring without specialist survey equipment.

Concrete Mix Ratio Predictor — ML Ensemble Model

Python · Scikit-Learn · Random Forest · Pandas

- Trained a MultiOutput Random Forest ensemble (300 estimators, max depth 12) on 712 real-world concrete mix design records to simultaneously predict fine and coarse aggregate-to-cement ratios, achieving $R^2 = 0.91 / 0.90$ and $MAE = 0.08 / 0.09$ on a held-out 20% test set.
- Engineered three physically grounded features, paste volume, fine fraction, and binder efficiency to augment raw mix parameters, improving model interpretability and predictive stability.
- Benchmarked four algorithms (Linear Regression, Decision Tree, Random Forest, XGBoost); confirmed Random Forest generalisation via 5-fold cross-validation (mean $R^2 = 0.89 \pm 0.03$), with fine fraction (importance: 0.31) and paste volume (0.18) identified as dominant predictors.
- Serialised the trained model using Joblib for integration into downstream web and mobile applications as a reusable inference module.

For more projects and details, visit [\(Link\)](#).

HONOURS AND AWARDS

Top 100 Africa Future Leaders — Class of 2025

2025

2025 Top 100 Africa Future Leaders Programme Selected from a competitive continental pool as one of 100 outstanding young African leaders demonstrating exceptional innovation and professional promise.

Vice-Chancellor's List — 37th & 38th Founders' Day

2022 & 2023

Federal University of Technology, Minna Recognised by the Vice-Chancellor on two consecutive occasions for outstanding academic performance, awarded to the highest-achieving students across the university.

Petroleum Technology Development Fund (PTDF) In-Country Scholarship

2021 – 2024

Petroleum Technology Development Fund, Federal Government of Nigeria Competitive national scholarship awarded to high-achieving engineering students across Nigerian universities, covering four consecutive years of studies.

Award of Excellence for Outstanding Academic Performance — Harmony Awards

2022

Nigerian Universities Engineering Students Association (NUESA), FUT Minna Recognised for exceptional academic and professional contributions within the engineering student community.

SOME CERTIFICATIONS AND ONLINE COURSES

AI & Machine Learning

Masterclass in Data Science using Python

CS221: Artificial Intelligence — Stanford

Python Data Structures and Algorithms

Software Engineering & Architecture

Software Architecture, Engineering and Development

Mastering WordPress Development and Monetisation

Analytics & Growth

Google Ads & Google Analytics 4

Google Search Console Mastery